# Thomas Putzer

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## Education

**Technical University of Munich** — Munich, Germany
*M.Sc. Informatics: Games Engineering, Current Grade: 1.2 (US GPA equivalent: ∼4.0/4.0)* — *Oct. 2025 – Ongoing*
***Exchange Semester: Tsinghua University (Beijing, PRC)*** — *Sept. 2026 – Jan. 2027*

**Technical University of Munich** — Munich, Germany
*B.Sc. Informatics: Games Engineering, Grade: 1.3 (US GPA equivalent: ∼4.0/4.0)* — *Oct. 2022 – Sep. 2025*
***Exchange Semester: Seoul National University (Seoul, South Korea)*** — *Sept. 2024 – Jan. 2025*

## Experience

**Working Student: Software Development** — Nov. 2025 – Aug. 2026
*VECTOR Informatik* — *Munich, Germany*
- Worked on an interactive editor for OpenDrive files built with Unity.
- Developed an import feature for GPX, GeoJSON, and CSV formats to convert waypoints into smooth splines.
- Developed a feature to connect two roads or lanes via G2-continuous splines using clothoids and arcs.

**Substitute Teacher** — Mar. 2023 & Apr. 2024
*Technical High School Bruneck (TFO Bruneck)* — *Bruneck, Italy*
- Taught *electrical engineering and electronics* (electric motors and transistors).
- Taught *automation* (WinSPS/Step7, Arduino, Python, and web technologies (HTML/CSS/JavaScript)).

## Projects

**PBR-3DGS** | *Python, PyTorch, CUDA* | github.com/PuuTzzA/PBR-3DGS — Apr. 2026 – Jul. 2026
- Adapted an inverse 3D Gaussian Splatting framework to use diffusion priors for material decomposition into albedo, normal, roughness, and metallic.
- Developed custom loss functions to successfully integrate diffusion priors.

**Global Logistics** | *Unity, HLSL, ShaderGraph* | github.com/mulsicpp/MasterGamesLab — Apr. 2026 – Jul. 2026
- Developed a real-time online multiplayer strategy game (Game Wiki).
- Implemented custom vertex shaders for map projection alongside a custom render pipeline for screen-space outlines and element selection.

**sdf-ui** | *JavaScript, GLSL, WebGL, HTML, CSS* | github.com/PuuTzzA/sdf-ui — Feb. 2026 – May. 2026
- Developed a dependency-free JavaScript and WebGL framework to dynamically transform HTML elements into 3D signed distance fields.

**SfM** | *C++, Ceres, OpenCV, Eigen, OpenMP, TBB, Blender* | github.com/PuuTzzA/SfM — Sep. 2025 – Jan. 2026
- Implemented a pipeline for Structure from Motion (SfM), including camera calibration.
- Programmed the 8-point algorithm for initial pose estimation and bundle adjustment using Ceres from scratch.
- Created a Python script to import the calculated camera poses and points into Blender.

**Bachelor Thesis** | *C++, Python, Mantaflow* | github.com/PuuTzzA/mantaflow — Apr. 2025 – Sep. 2025
- *Higher-Order Mass- and Momentum-Conserving Advection for Fluid Simulations with Large Time Steps*
- Implemented and refined a custom mass- and momentum-conserving advection algorithm.
- Evaluated and compared several different advection schemes.

## Awards and Stipends

**Deutschlandstipendium** | *Merit-based German national stipend for top-performing students* — 2023 & 2024 & 2025
**best.in.tum** | *Award from the Department of Informatics at TUM for the best 2% of its students.* — 2024

## Skills

**Programming Languages**: C#, C/C++, GLSL, Python, Java, JavaScript, HTML/CSS
**Libraries & Frameworks**: PyTorch, Ceres Solver, OpenCV, Eigen, OpenMP, TBB, Mantaflow, CUDA
**Tools**: Unity, Blender, Git, Linux, DaVinci Resolve, Django, VS Code, Rider, IntelliJ, Arduino, WinSPS-S7
**Languages**: German (Native), English (Fluent/C1), Italian (Upper-Int./B2), Korean (Int./B1), Chinese (Basic/A2)